

Request for Budgetary Technical and Commercial Proposal No. OVE75-RFQ-13

Subject: Polymer-concrete electrowinning cells, 180 pcs.

KVANT LLC (Russian Federation), within the Basic Engineering (FEED) stage of the construction project of a copper production plant — a roasting-leaching-electrowinning complex with a capacity of 75,000 tonnes of copper cathodes per year — kindly requests your budgetary technical and commercial proposal (budgetary quotation) for the item above. Project site: an operating metallurgical site in north-west Russia.

Summary of the item: Inside dimensions 8485 × 1155 × 1650 mm; 84 cathodes / 85 anodes; integrated rim exhaust and removable hood; bottom sloped to the centre with a bottom drain plug. Technical requirements are given in Annex 1 to this request.

Please include in your quotation:

1. budget price quoted separately on EXW (manufacturer's works) and DDP project site — an operating metallurgical site in north-west Russia (Incoterms 2020) terms, stating the currency and the validity period of the offer;
2. manufacturing lead time and estimated delivery time (weeks from order / advance payment);
3. overall dimensions and weights of the equipment unit and of the largest shipping blocks (for logistics studies);
4. reference list of similar supplies: product, process duty and parameters, year of supply, country;
5. materials of parts in contact with the process medium, and the scope of supply (instrumentation, spare parts, documentation);
6. utility consumption and installation requirements (if available).

This is a budgetary enquiry at the Basic Engineering (FEED) stage: it does not constitute an offer and creates no obligations for either party; detailed technical data sheets will be issued at the next project stage.

Reply deadline: we kindly ask you to submit your quotation within 3 (three) weeks from receipt of this request to stepan@kvantpro.com. Please confirm receipt of this request.

Confidentiality. This request and the attached technical materials are provided solely for the preparation of the quotation, are confidential, and shall not be disclosed to third parties, copied or published without the prior written consent of KVANT LLC. Submission of a quotation creates no legal obligations for either party.

Contact person: Stepan Kharkov, KVANT LLC, stepan@kvantpro.com.

Annex 1: Technical requirements for the requested item.

Annex 1. Technical Requirements

To Request No. OVE75-RFQ-13 · The values follow the Customer's input data and the Customer's Terms of Reference; Basic Engineering (FEED) stage, subject to refinement at the next project stage.

RFQ item	Polymer-concrete electrowinning cells, 180 pcs
Summary	Inside dimensions 8485 × 1155 × 1650 mm; 84 cathodes / 85 anodes; integrated rim exhaust and removable hood; bottom sloped to the centre with a bottom drain plug
Specific	Confirm the cell design and material (monolithic vinyl-

requirements	ester composite) and the integration of the rim exhaust and the easily removable hood
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Equipment covered by this request:

1. Electrowinning cell Electrowinning · item 4.2.3

Type / model	Polymer-concrete cells of monolithic vinyl-ester composite; design compliance to be confirmed in the quotation
Duty	Copper electrowinning from solutions
Key parameters	Actual current $I = 42.45$ kA; design current density 280.0 A/m ² , actual 252.7 A/m ² ; output 49.49 kg/h per cell, $8,066.1$ kg/h in total
Quantity	180 (2 sections × 90); calculated 163 pcs, margin factor 1.104
Materials	Polymer concrete (monolithic vinyl-ester composite)
Dimensions, hold-up	Inside dimensions $8485 \times 1155 \times 1650$ mm (L × W × H); 84 cathodes, 85 anodes
Additional requirements	The design is integrated with the aspiration system: rim exhaust + easily removable hood. Bottom sloped to the centre, with a bottom plug for slime discharge. Lead slime clean-out normally not more than once a year.

2. Cell aspiration enclosures Electrowinning

Type / model	Rim exhaust + easily removable hood
Duty	Local aspiration of the cells preventing the dew point from being reached under the hoods
Key parameters	Periodic water flushing of the collectors against accumulation of dry salts; flushing solutions drained to a buffer tank or to the scrubber feed
Quantity	180 sets
Materials	Corrosion-resistant design

The quotation must confirm: process medium, temperature, pressure, materials of wetted parts, capacity, dimensions and weight, power consumption, manufacturing lead time, delivery terms.

预算性技术商务报价询价函 编号 OVE75-RFQ-13

询价主题：聚合物混凝土电积槽，180 台。

KVANT LLC（俄罗斯）正在开展一座铜冶炼厂建设项目的基础设计（FEED）——“焙烧—浸出—电积”联合工艺，年产阴极铜 75,000 吨——特此请贵方就上述设备提交预算性技术商务报价。项目场地：俄罗斯西北部一座现役冶金厂区。

设备简述：内部尺寸 8485 × 1155 × 1650 mm；84 片阴极 / 85 片阳极；集成槽沿排风及可拆卸罩盖；槽底向中心倾斜并设底部排放塞。技术要求见本询价函附件 1。

请在报价中包含以下内容：

1. 预算价格——分别按 EXW（制造厂交货）和 DDP 项目场地（俄罗斯西北部现役冶金厂区）条款报价（Incoterms 2020），并注明币种及报价有效期；
2. 制造周期及预计交货期（自下单/预付款之日起，按周计）；
3. 外形尺寸与重量：单台设备及最大发运件的尺寸和重量（用于物流研究）；
4. 类似供货业绩表：产品、工艺介质及参数、供货年份、国家；
5. 与工艺介质接触部件的材质，以及供货范围（仪表、备品备件、文件资料）；
6. 公用工程消耗及安装要求（如有资料）。

本询价为基础设计（FEED）阶段的预算性询价：不构成要约，对双方均不产生任何义务；详细技术数据表将在项目下一阶段发出。

回复期限：请自收到本询价函之日起 3（三）周内将报价发送至 stepan@kvantpro.com。请确认收到本询价函。

保密条款：本询价函及所附技术资料仅供编制报价使用，属于保密信息，未经 KVANT LLC 事先书面同意，不得向第三方披露、复制或发表。提交报价不构成双方的法律义务。

联系人：Stepan Kharkov, KVANT LLC, stepan@kvantpro.com。

附件 1：本询价设备的技术要求。

附件 1. 技术要求

对应询价函编号 OVE75-RFQ-13 · 参数依据业主原始数据及设计任务书；基础设计（FEED）阶段，下一阶段可能进一步细化。

询价设备	聚合物混凝土电积槽，180 台
简述	内部尺寸 8485 × 1155 × 1650 mm；84 片阴极 / 85 片阳极；集成槽沿排风及可拆卸罩盖；槽底向中心倾斜并设底部排放塞
特殊要求	需确认槽体结构与材质（整体浇注乙烯基酯复合材料），以及槽沿排风与轻便可拆罩盖的集成

本询价涵盖的设备清单：

1. 电积槽 电积 · 位号 4.2.3

型式／型号	整体浇注乙烯基酯复合材料的聚合物混凝土电解槽；结构符合性在报价中确认
用途	从溶液中电积铜
主要参数	实际电流 $I = 42.45 \text{ kA}$ ；设计电流密度 280.0 A/m^2 ，实际

	252.7 A/m ² ；产能：每槽 49.49 kg/h，合计 8,066.1 kg/h
数量	180 台（2 系列 × 90）；计算值 163 台，裕量系数 1.104
材质	聚合物混凝土（整体浇注乙烯基酯复合材料）
尺寸、持液量	内部尺寸 8485 × 1155 × 1650 mm（长×宽×高）；84 片阴极、85 片阳极
附加要求	结构与通风除尘系统一体化：槽沿排风+轻便可拆罩盖。槽底向中心倾斜，设底部排放塞用于排出阳极泥。铅泥清理通常每年不超过 1 次。

2. 电解槽通风除尘罩 电积

型式/型号	槽沿排风 + 轻便可拆罩盖
用途	电解槽局部排风，防止罩盖下达到露点
主要参数	定期用水冲洗集气管以防干盐积聚；冲洗液排入缓冲罐或洗涤器给液系统
数量	180 套
材质	耐腐蚀材质

报价中必须确认：工艺介质、温度、压力、接触部件材质、生产能力、尺寸与重量、能耗、制造周期、交货条件。